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Quentin Rebergue - Graduate Software Engineer

Profile

Looking for a full-time position in software engineering with a focus on embedded systems, low-level, hardware interface, Linux. MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded.

Fluent in French and English (TOEIC 915, C1). Interests: watchmaking, music, hiking/running , cooking.

Work Experience

- **Software Engineer Intern - Platform team, ARISTA**, Dublin 02/2026 – 08/2026

  Implemented non-zero reset value management for ASIC Physical Table registers across the chip-agnostic Strata infrastructure (SDK parsing, code generation, DMA) in python and C++. Won 1st place at Arista's Hackathon by building an AI agent optimization tool (70% token reduction on internal codebases). Approved for production development.

- **Head of Teaching Assistants - YAKA** / **ACU Program, EPITA**, Kremlin-Bicetre 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, malloc, POSIX shell) for 700+ 3rd-year students across 5 campuses. Implemented 1-on-1 code review mentorship model, improving TA-to-student ratio from 1:30 to 1:20.

School Projects

- **ARM project** – Built an STM32F429ZI embedded system acting as a UART command server using libgapcom (Protobuf over UART). Implemented a 4-level log subsystem and an MPU6050 gyroscope driver over I2C with FIFO reads.Team of 2.
- **RFID Driver** – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.
- **Synchronous Real-Time Systems** – Built a Rate Monotonic scheduler in Heptagon, a dataflow language that compiles nodes to deterministic C step functions with no heap allocation, and deployed it to Raspberry Pi 3 via UART bootloader.
- **GPU-Accelerated Video Analysis System (IRGPU)** – Built a real-time motion detection pipeline in C++/CUDA (reservoir sampling, morphological ops, hysteresis thresholding). Achieved 20x CPU speedup through iterative CUDA optimization profiled with Nsight Compute. Team of 4.
- **My Docker** – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.

Education

- **MSc Computer Engineering, Low-level & Systems (GISTRE)**, EPITA 2020 – 2026
- **Preparatory Class (PTSI) in Engineering Science**, Lycée Jules Ferry Versailles 2019 – 2020

Technical Expertise

- **Programming & Languages:** ADA, ASM, Bash, C, C++, C++20, CUDA, Python, Rust, Shell.
- **Embedded & Hardware:** ASIC interfacing, FPGA, I2C, Raspberry Pi, STM32, UART, SPI.
- **Other** / **DevOps:** GDB, Git, Security, monitoring, optimisation, testing, valgrind.
- **Systems:** Bootloader, DMA, Drivers, Kernel, Linux, Multithreading, Virtualization, low-latency, real-time.